

Electric Fields

Chapter 22 - P.H.W - 10th Edition

Electric Fields

(8)

from our previous lectures:

① We talk about electric charges and forces and we know that an electrostatic force acts on particle 1 due to the presence of particle 2. We also know the force direction and with some data, we can calculate the force magnitude.

Question? 1. How does particle 1 'know' of the presence of particle 2?

2. Since particles do not touch, how can particle 2 push on particle 1.

How can there be such an action at a distance?

Our purpose in this chapter is to provide to answer of these questions on electric force at a distance.

The Electric Field

A lot of different fields are used in science and engineering.

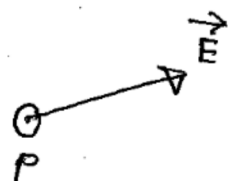
→ Temperature field (distribution of Temperature)

→ pressure field (distribution of pressure)

Such fields are examples of scalar fields, because temperature and pressure are scalar quantities, having only magnitudes and not directions.

In contrast!, an electric field is a vector field because it is responsible for conveying the information for a force, which involves both magnitude and direction.

This field consists of a distribution of electric field vector $\vec{E}$, one for each point in the space around a charged object.



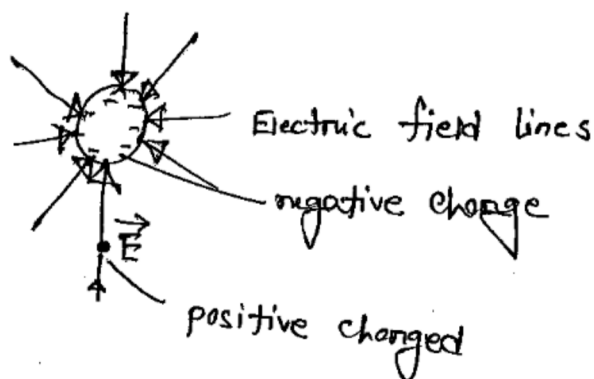
Electric field at point P.

At 'P' we place a particle with a small positive charge q_0 , called a test charge because we use it to test the field. (test charge to be small so that it does not disturb the object charge distribution), we then measure the electrostatic force

$\vec{F}$ that acts on the test charge. The electric field at that point is then $\vec{E} = \frac{\vec{F}}{q_0}$ (Electric field)

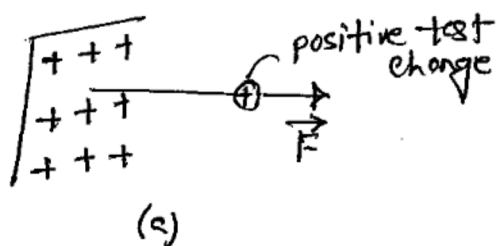
Electric field lines

Michael Faraday, who introduced the idea of electric fields in the 19th Century, he envisioned lines, now called electric field lines, in the space around any given charged particle or object.

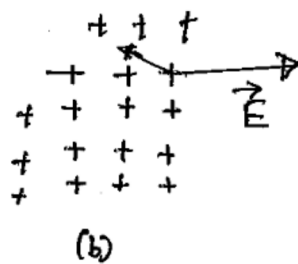


If we place a positive test charge at any point near the sphere, we find that an electrostatic force pulls on it toward the center of the sphere.

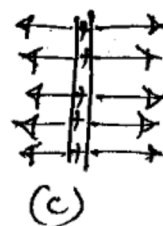
Thus at every point around the sphere, an electric field vector points radially inward toward the sphere.



The force on a positive test charge near a very large, nonconducting sheet with uniform positive charge on one side.



The electric field vector $\vec{E}$ at the test charge's location, and the nearby electric field lines, extending away from the sheet.



Electric field lines extend away from positive charge and towards negative charge.

The electric field due to a charged particle

The electric field due to a point charge

To find the electric field due to a charged particle (often called a point charge), we place a positive test charge at any point near the particle, at distance r . From Coulomb's law the force on the test charge due to the particle with charge q is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q q_0}{r^2} \hat{r} \quad \left[\begin{array}{l} \text{The direction of } \vec{F} \text{ is} \\ \text{directly away from the} \\ \text{particle if } q \text{ is positive} \\ \text{and directly towards it} \\ \text{if } q \text{ is negative} \end{array} \right]$$

Now, electric field set up by the

particle as, $\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$ (Charged particle)

(Test charge)

Let's think through the direction again.

The direction of $\vec{E}$ matches that of the force on the positive test charge: directly away from the point charge if q is positive and directly toward if q is negative.

We can find the magnitude at any given distance r by unvectoring $E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}$ (Charged particle)

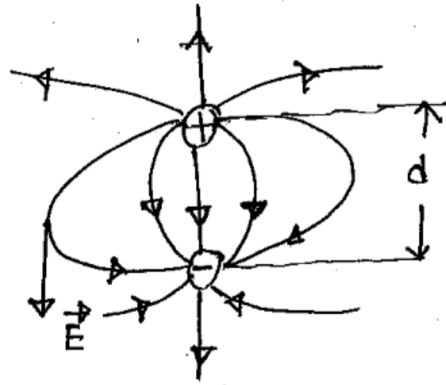
(To avoid charge notation)

$$\vec{F}_0 = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots \quad (\text{forces obey the principle of superposition})$$

$$\vec{E} = \frac{\vec{F}_1}{q_0} + \frac{\vec{F}_2}{q_0} + \frac{\vec{F}_3}{q_0} + \dots = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$$

The Electric Field Due to A Dipole

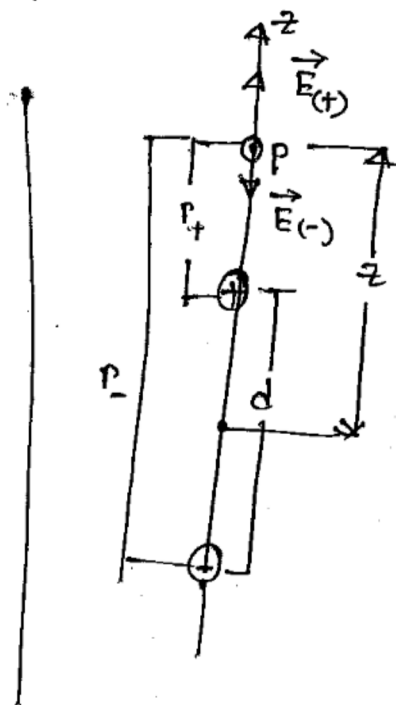
(1)



d - along the dipole axis,
let's consider it z.
(with symmetry).

fig: The pattern of electric field lines around an electric dipole, with an electric field vector $\vec{E}$ shown at one point (tangent to the field line through that point).

We will find out Electric field $\vec{E}$ at an arbitrary point P along the dipole axis,



The magnitude of the net field at P as

$$\begin{aligned} E &= E_{(+)} - E_{(-)} \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{r_{(+)}^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{r_{(-)}^2} \\ &= \frac{q}{4\pi\epsilon_0 (z - \frac{d}{2})^2} - \frac{q}{4\pi\epsilon_0 (z + \frac{d}{2})^2} \end{aligned}$$

$$= \frac{q}{4\pi\epsilon_0 z^2} \left(\frac{1}{(1 - \frac{d}{2z})^2} - \frac{1}{(1 + \frac{d}{2z})^2} \right) \quad \text{--- (1)}$$

$$\begin{aligned} &= \frac{q}{4\pi\epsilon_0 z^2} \frac{2d/z}{(1 - (\frac{d}{2z})^2)^2} \\ &= \frac{q}{4\pi\epsilon_0 z^3} \frac{d}{(1 - (\frac{d}{2z})^2)^2} \quad \text{--- (2)} \end{aligned}$$

We are usually interested in the electrical effect of a dipole only at a distance that are large compared with the dimensions of the dipole -

limit such that $z \gg d$.

eqn (2) $\Rightarrow$ becomes;
$$E = \frac{1}{4\pi\epsilon_0} \frac{qd}{z^3}$$

The product qd , which involves the two intrinsic properties q and d of the dipole, is the magnitude P of a vector quantity known as the electric dipole moment $\vec{P}$ of the dipole. $P \rightarrow \text{unit} \rightarrow (\text{Coulomb-metres})$

$$E = \frac{1}{4\pi\epsilon_0} \frac{P}{z^3} \text{ (electric dipole).}$$

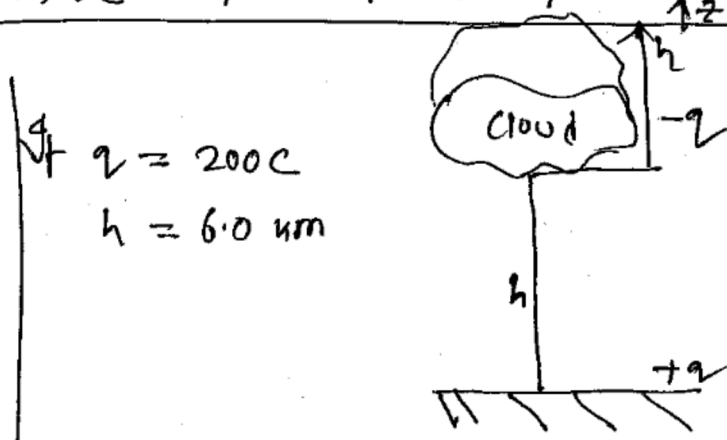
We use direction of $\vec{P}$ to specify the orientation of a dipole.

Sample problem 22.02

Electric dipole and atmospheric sprites

$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0} \frac{q(2h)}{z^3} \\ &= \frac{1}{4\pi\epsilon_0} \frac{(200\text{C})(2)(6.0 \times 10^3\text{m})}{(30 \times 10^3\text{m})^3} \\ &= 1.6 \times 10^3 \text{ N/C} \end{aligned}$$

For z_2 , $E = 2 \times 10^2 \text{ N/C}$



$\rightarrow$ We will discuss later: An electric field exceeds a certain critical value E_c , the field can pull electrons out of atoms (ionized the atoms) and then the freed electrons can run into other atoms, causing these atoms to emit light.

2

The Electric field Due to a Line of Charge

The charge of an extended object is often conveyed in terms of a charge density rather than the total charge.

First Q! We have an equation for the electric field, now we have line of charge?

→ dq - element

2nd Q! Each charge element dq and

write an expression for its contribution of electric field $d\vec{E}$.

But how to sum up for a line of charge's contribution to get net field at P?

→ Split vectors in component

3rd Q! A huge number of dq , thus a huge number of $d\vec{E}$ component to add up.

→ integration.

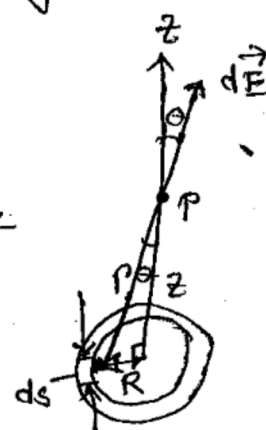
Now, elementary charge.

~~First Q~~ $dq = \lambda ds$ — (1)

Where, $\lambda \rightarrow$ linear charge density.

$ds \rightarrow$ Arc length.

Electric charge		
	Symbol	SI unit
Charge	q	C
Linear charge density	λ	C/m
Surface charge density	σ	C/m ²
Volume charge density	ρ	C/m ³



A ring of uniform positive charge.

An Element's field dq charge element sets up the differential electric field $d\vec{E}$ at P, at distance r from the element, (figure above).

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2} \quad \text{--- (1)}$$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{(z^2 + R^2)} \quad \text{--- (iii)}$$

Because every charge element has the same charge and the same distances from point P, eq (iii) gives the field magnitude contributed by each of them.

Each $\vec{dE}$ lies at angle θ to the central axis — and thus has components perpendicular and parallel to that axis.

Two perpendicular components cancel each other. All around the ring, this cancellation occurs for every charge element and its symmetric partner on the opposite side of the ring. So, we can neglect all the perpendicular components.

→ All the remaining components are in the positive direction of the z axis, we can just add them as scalars.

$$dE \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{z^2 + R^2} \times \frac{z}{(z^2 + R^2)^{1/2}}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{z \lambda}{(z^2 + R^2)^{3/2}} ds \quad \text{--- (iv)}$$

$$E = \int dE \cos \theta \Rightarrow \frac{z \lambda}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} \int_0^{2\pi R} ds$$

$$\Rightarrow \frac{\lambda \cdot z (2\pi R)}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}}$$

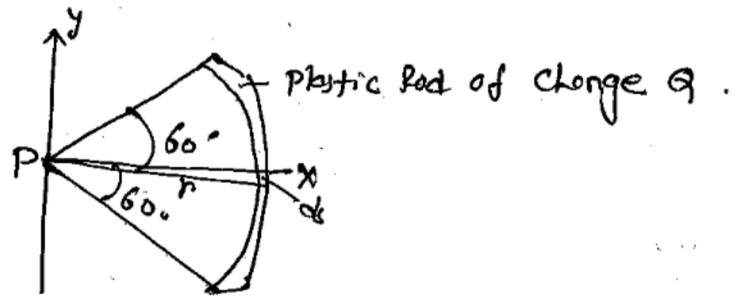
$$= \frac{qz}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} \quad (\text{charged ring})$$

Where $z \gg R$
 Ring looks like a point charge $= \left(\frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \right)$ (charged ring at large dis.)

Sample problem

Electric field of a charged circular rod.

(21)



$$dq = \lambda ds$$

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2}$$

$$dE_x = dE \cos\theta \Rightarrow \frac{1}{4\pi\epsilon_0} \frac{\lambda}{r^2} \cos\theta ds \Rightarrow \frac{1}{4\pi\epsilon_0} \frac{\lambda}{r^2} \cos\theta (r d\theta)$$

$$E = \int dE_x = \int_{-60^\circ}^{60^\circ} \frac{1}{4\pi\epsilon_0} \frac{\lambda}{r^2} \cos\theta r d\theta$$

$$= \frac{\lambda}{4\pi\epsilon_0 r} \int_{-60^\circ}^{60^\circ} \cos\theta d\theta$$

$$= \frac{\lambda}{4\pi\epsilon_0 r} [\sin\theta]_{-60^\circ}^{60^\circ}$$

$$= \frac{1.73\lambda}{4\pi\epsilon_0 r}$$

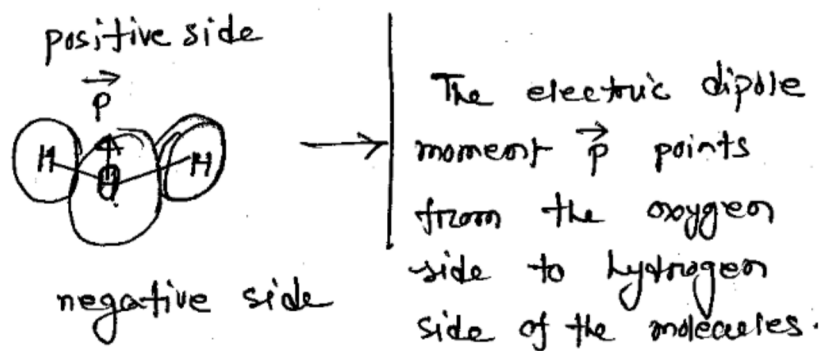
$$\lambda = \frac{\text{Charge}}{\text{length}} \Rightarrow \frac{Q}{2\pi r/3} \Rightarrow \frac{0.477Q}{r}$$

$$E = \frac{0.83Q}{4\pi\epsilon_0 r^2}$$

$$E \text{ in unit vector } \vec{E} = \frac{0.83Q}{4\pi\epsilon_0 r^2} \hat{i}$$

A Dipole in An Electric Field

(4)



We have defined the electric dipole moment $\vec{p}$ of an electric dipole to be a vector that points from the negative to positive end of the dipole.

② The behaviour of a dipole in a uniform external electric field $\vec{E}$ can be described completely in terms of the two vectors $\vec{E}$ and $\vec{p}$, with no need of any details about the dipole's structure.

□ In a water molecule, the two hydrogen atoms and the oxygen atom do not lie on a straight line but form an angle of about 105° (see above figure). As a result, the molecule has a definite 'oxygen side' and 'hydrogen side'. Moreover, the 10 electrons of the molecule tend to remain closer to the oxygen nucleus than to the hydrogen nuclei. This makes the oxygen side of the molecule slightly more negative than the hydrogen side and creates an electric dipole moment $\vec{p}$ that points along the symmetry axis of the molecule.

(7)

If the water molecule is placed in an external electric field, it behaves as would be expected of the more abstract electric dipole.

□ We assume that the dipole is a rigid structure that consists of two centers of opposite charge, each of magnitude q , separated by a distance d . The dipole moment $\vec{p}$ makes an angle θ with field $\vec{E}$.

□ Electrostatic forces act on the charged ends of the dipole. Because the electric field is uniform, those forces act in opposite directions and with the same magnitude $[F = qE]$. Because the field is uniform, the net force on the dipole from the field is zero and the center of mass of the dipole does not move.

However, the forces on the charged ends do produce a net torque $\vec{\tau}$ on the dipole about its center of mass. The center of mass lies on the line connecting the charged ends, at some distance x from one end and thus a distance $d - x$ from the other end.

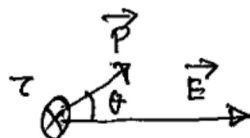
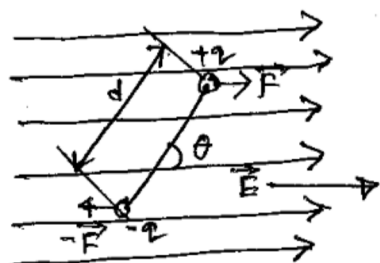
$$\tau = r F \sin\theta \quad (\text{Phy-107})$$

$$\tau = Fx \sin\theta + F(d-x) \sin\theta = Fd \sin\theta$$

$$\text{As } p = qd, \quad \tau = F \frac{p}{q} \sin\theta$$

$$\tau = qE \frac{p}{q} \sin\theta = pE \sin\theta$$

$$\vec{\tau} = \vec{p} \times \vec{E} \quad (\text{Torque on dipole})$$



Sign of the torque: We can represent a torque that gives rise to a clockwise rotation by including a minus sign with the magnitude of the torque.

$$\tau = -PE \sin \theta$$

Potential Energy of an Electric Dipole

Potential energy can be associated with the orientation of an electric dipole in an electric field.

① The dipole has its least potential energy when it is in its equilibrium orientation (When its moment $\vec{p}$ is lined up with the field $\vec{E}$) so, $\vec{\tau} = \vec{p} \times \vec{E} = 0$

② It has greater potential energy in all other orientation.

We can find the potential energy U of the dipole at any other value of θ by calculating the work W done by the field on the dipole when the dipole is rotated to that value of θ from 90° .

$$\begin{aligned} W &= \int \tau d\theta \\ U &= -W = - \int_{90}^{\theta} \tau d\theta = \int_{90}^{\theta} pE \sin \theta d\theta \\ &= -pE \cos \theta \\ &= -\vec{p} \cdot \vec{E} \quad (\text{potential energy of a dipole}) \end{aligned}$$

Sample Problem 22.05. Torque and Energy of an electric dipole in an electric field. (16)

- (A) H_2O (neutral) in its vapor state has an electric dipole moment of magnitude $6.2 \times 10^{-30} \text{ C}\cdot\text{m}$.

How far apart are the molecule's centers of positive and negative charge?

$$P = qd = (10e)(d)$$

$$q = \frac{P}{10e} = \frac{6.2 \times 10^{-30} \text{ C}\cdot\text{m}}{(10)(1.60 \times 10^{-19} \text{ C})} = 3.9 \times 10^{-12} \text{ m}$$
$$= 3.9 \text{ pm}$$

This distance is not only small, but it is also actually smaller than the radius of a hydrogen atom.

- (B) If the molecule is placed in an electric field of $1.5 \times 10^4 \text{ N/C}$, what maximum torque can the field exert on it?

* The torque on a dipole is maximum when the angle θ between $\vec{p}$ and $\vec{E}$ is 90° .

Substituting $\theta = 90^\circ$,

$$\begin{aligned}\tau &= pE \sin \theta \\ &= (6.2 \times 10^{-30} \text{ C}\cdot\text{m})(1.5 \times 10^4 \text{ N/C})(\sin 90^\circ) \\ &= 9.3 \times 10^{-26} \text{ N}\cdot\text{m}\end{aligned}$$

- (C) How much work must an external agent, in order to change the molecule's potential energy due to a change in orientation?

$$W_a = U_{180^\circ} - U_0$$

② How much work must an external agent do to rotate this molecule by 180° in this field, starting from its fully aligned position, for which $\theta = 0$?

$$W_a = U_{180^\circ} - U_0$$

$$= (-PE \cos 180^\circ) - (-PE \cos 0)$$

$$= 2PE = (2)(6.2 \times 10^{-30} \text{ C.m})(1.5 \times 10^4 \text{ N/C})$$

$$= 1.9 \times 10^{-25} \text{ J}$$

Answer:

A point charge in an Electric field

$$\vec{F} = q\vec{E}$$

The electrostatic force $\vec{F}$ acting on a charged particle located in an external electric field $\vec{E}$ has the direction of $\vec{E}$ if the charge q of the particle is positive and has the opposite direction if q is negative.

Electrical Breakdown and Sparking

If the magnitude of an electric field in air exceeds a certain critical value E_c , the air undergoes electrical breakdown, a process whereby the field removes electron from the atoms in the air.

Measuring elementary charge

American Physicist Robert A. Millikan in 1910-193 measured elementary charge e .

Millikan discovered that the value of q ~~where~~ were always given by $q = ne$, for $n = 0, \pm 1, \pm 2, \pm 3, \dots$ in which e turned out to be the fundamental constant we call the elementary charge

$$e = 1.60 \times 10^{-19} \text{ C.}$$

Due to this work he got Nobel Prize in Physics, -1923.